Q.2

Assignment 5

LC	Symbols	Ins	Ref	obj code
0000	copy	Start	0000	No obj code
0000		LDS	#3	6D0003
0003		LDT	#300	75012C
0006		LDX	#0	050000
000C		LDA	ALPHA,X	03A00D
000F		ADD	BETA,X	1BA36
0012		STA	GAMMA,X	0FA38B
0015		ADDR	S,X	9041
0017		COMPR	X,T	A015
0019		JLT	ADDLP	3B2FF0
001C	ALPHA	RESW	100	No obj
0148	BETA	RESW	100	No obj
0274	GAMMA	RESW	100	No obj
03A0		END		No obj

Symbol TAGle

Symbol	Address
ADDLP	000C
ALPHA	001C
BETA	0148
GAMMA	03A0

H.COPYXX.000000.0003A0
T.000000.19.6D0003.75012C.050000.03A00D.1BA136.0FA38B.9041.A015.3B2FF0
E.000000

OPcode	n	i	x	b	p	e	disp
6C 01101	0	1	0	0	0	0	003 0000 0000 0011
LDT #300 74 011101	0	1	0	0	0	0	12C 0001 0010 1100
LDX #0 04 0000 01	0	1	0	0	0	0	000 0000 0000 0000
LAD ALPHA, x 00 0000 00	1	1	1	0	1	0	00D 0000 0000 1101 D disp = TA − PC 001C − 000F = 000D
ADD BETA, x 10 0001 10	1	1	1	0	1	0	136 0001 0011 0110 disp = TA − PC 0140 − 0012 = 0136
STA GTAMMA, x 0C 0000 11	1	1	1	0	1	0	38B 0011 1000 1011 disp = TA − PC 03A0 − 0015 = 38B

ADDR S, X 90 S=4 X=1

1001 0000

r_1 4 r_2 1

0100 0001

COMPR X, T X=1 T=1

A0 r_1 1 r_2 5

1010 0000 0001 0101

JLT ADDLP

38

0011 10

n i x b p e

1 1 0 0 1 0

disp

FF0

1111 1111

000

disp = TA − PC

000C − 001C = FF0

LC	Symbol	Ins	Ref	Obj code
0000		start	0000	No obj
0000	FIRST	STL	RETADR	17 2020
0003		LDB	#LENGTH	69 2020
0006		BASE	LENGTH	No obj
0006	CLOOP	+JSUB	RDREC	4B101036
000A		LDA	LENGTH	032026
000D		COMP	#0	290000
0010		JEQ	ENDFIL	332007
0013		+JSUB	WRREC	
0017		J	CLOOP	3F 2FEC
001A	ENDFIL	LDA	EOF	032010
001D		STA	BUFFER	0F 2016
0020		LDA	#3	010003
0023		STA	LENGTH	0F 2000
0026		+JSUB	WRREC	
002A		J	@RETADR	3E2003
002D	EOF	BYTE	C'EOF'	45 4F 46
0030	RETADR	RESW	1	No obj
0033	LENGTH	RESW	1	No obj
0036	BUFFER	RESB	4096	No obj
1036	RDREC	CLEAR	X	B410
1038		CLEAR	A	B400
103A		CLEAR	S	B440
103C		+LDT	#4096	77101000
1040	RLOOP	TD	INPUT	E32019
1043		JEQ	RLOOP	332FFA
1046		RD	INPUT	DB2013
1049		COMPR	A,S	A004
104B		JEQ	EXIT	332008
104E		STCH	BUFFER,X	57C003
1051		TIXR	T	B850
1053		JLT	RLOOP	3B2FEA
1056	EXIT	STX	LENGTH	134000
105C	INPUT			

OPcode	n	i	x	b	p	e	disp
LDB #length 1+)68 0110 10	0	1	0	0	1	0	02D disp = TA − PC 0000 0010 1101 0033 − 0006 02D
COMP, #0, 28 0010 10	0	1	0	0	0	0	000 0000 0000 0000
+JSUB, WRREC, 48 0100 10 +2	1	1	0	0	0	1	~~disp~~ Address
LDA #3 00 0000 00	0	1	0	0	0	0	003 0000 0000 0011
+JSUB WRREC, 48 0100 10	1	1	0	0	0	1	Address
BYTE C'EOF' 4 54646							
CLEARX x=1, 04 1011 0100	r1 r2 1 0 0001 0000						
+LDT #4096, 74 0111 01	1	1	0	0	0	1	Address → 01000 0000 0001 0000 0000 0000
COMPR A,S A=0 S=4 A0 10100000	r1=0 r2=4 0000 0100						
STCH BUFFER,X 54 0101 01	1	1	1	1	0	0	disp = TA − PC 003 0036 − 0033 = 003 0000 0000 0011

TD INPUT E32019 disp = TA − PC
019 = TA − 1043

TA = 105C

RD INPUT DB2013

disp = TA − PC
013 = TA − 1049

TA = 105C

Q.3

H. XSTART. 000000. 001056

T. 000000. 1D17202D. 69202D. 4B101036. 032026. 8bit. 290000. 332007. 3F2FEC. 032010

T. 000010. 1D. of 2016... 8-bit. 3E2003. 454F460

T. 001036. 1D. B410. B400. B410. 77101000. E32019. 332FFA. DB2013. A004. 332008. 570003. B850

T. 001053. 0C. 3B2FEA. 134000

M. 000007. 05
M. 000014. 05
M. 000027. 05
M. 001036. 05

E. 000000